Companion — WebLinked user guide

This module drives WebLinked over its HTTP control API from a Stream Deck or any other Companion surface: graphics, outputs, formats, and the health numbers that tell you whether a page is keeping up.

The README covers installing the module. This guide is about building a surface with it.

> **Before you rely on this: this module polls.** WebLinked has no push channel for state, so every feedback and variable can be up to one poll interval behind what's really happening. Don't treat any of it as instant.
>
> This module was built with AI assistance, directed and reviewed by a human author.

Connecting

`127.0.0.1:7654` by default. If WebLinked was started with `--bind 0.0.0.0` it should have been started with `--token` as well, so put that token in the config. **There's no TLS**, so keep it on a show network you trust.

Finding an instance

WebLinked 0.8.0 and later advertise themselves over mDNS, so the config panel gives you a **list to pick from** instead of an address to type. Pick one and the host and port fields disappear, because they're ignored from then on.

"Manual" is always there, and it's the answer whenever discovery comes up empty. That happens for four reasons, roughly in order of how often each one's to blame:

- **The instance is on the default loopback bind.** It won't advertise on purpose (nothing off its machine could reach the address it would publish) and says so in its log. Start it with `--bind 0.0.0.0`.
- **Companion is on another subnet.** mDNS doesn't cross a router.
- **Multicast is filtered**, which is common on managed venue networks.
- **The instance is older than 0.8.0**, or was started with `--no-mdns`.

None of these stop the module working. They only stop it filling the address in for you.

Run JavaScript is the high-leverage action

If a graphic defines its own functions, you can drive it with no integration work at all:

```
lowerThird.show('Anna Kowalski', 'Head of Sound')
setScore('home', 3)
```

That's the action to build a show around. Anything the page can do, a button can do, and nobody has to write a protocol for it first.

Enabled is not running

- **Enabled** is what you've asked for.
- **Running** means the device actually opened and is putting frames out.

An output can be enabled and not running: a DeckLink claimed by another application, say, or a format the card won't take. That's why the generated output presets light on **running**, and you should keep them that way.

Disabling an output stops the device and frees it for another application. It's how you hand a card to something else without quitting WebLinked.

The health numbers, and how to read each one

Feedback	Read it as
Dropped ticks	Should be **0**. Anything else means the clock fell behind.
Repeated frames	High is **normal** for a static graphic. Rising on an animated one means the page is too slow — set the threshold for what *this* graphic should be doing.
Console errors	The page is throwing. The graphic is probably wrong.
Popup attempts	The page tried to open a window; it is usually about to misbehave.
Audio underruns	A few at start-up are normal. A **rising** count is not.

Repeated frames is the one that catches people out. The same number means "fine" on a static lower third and "this page can't keep up" on an animation, so the threshold is per graphic.

Background is per output

Each output puts the page over either **its own transparency** (a keyed SDI fill, or NDI with alpha) or **a flat colour**, for a switcher that's only got a chroma keyer.

One browser paint serves both, so the same graphic can go out as a key down one path and over green down another, without rendering it twice.

A change takes effect on the next tick **without restarting the output**, so it won't drop frames on air.

The source selector

WebLinked can run several pipelines in one process. Every per-source action and feedback has a **Source** dropdown that defaults to *Primary*, which sends no source parameter at all, the way WebLinked expects.

Leave it alone unless the rig runs more than one pipeline.

When Format returns 409

A 409 means the format did change, but an output couldn't reopen at it.

Fire **Diagnostics: log the current state** and read `outputs[].error`. That's the only place the actual reason shows up.

Building a surface that fails safe

1. **WebLinked is connected** on any page that shows output colour. While WebLinked can't be reached, every other feedback keeps working from the *last known* state, so a brief blip doesn't blank the page, but it also means a stale green looks the same as a live one.
2. **Output tiles that light on running**, not enabled.
3. **Dropped ticks somewhere you can see it.** It should be zero, so any colour at all is worth noticing.
4. **Repeated-frames thresholds set per graphic**, not globally.

Troubleshooting

Symptom	Cause
Discovery list is empty	One of the four reasons above. Use Manual; the module still works.
Output shows enabled but nothing is on air	Enabled is intent; check running. The device may be claimed elsewhere.
Format action returned 409	The format changed but an output could not reopen. Log the state and read `outputs[].error`.
Repeated frames alarming on a static graphic	Normal. Set the threshold for what this graphic does.
Everything on the page is stale	The module polls. Also check *WebLinked is connected*.
Run JavaScript does nothing	The function does not exist on the page, or the page is throwing — check the console-errors feedback.

See also

- README: installing, and the full list of actions, feedbacks and variables
- `companion/HELP.md`: the same material, in Companion's help panel